ARC | System Architecture

Sensor acquisition and V5 communication

01 SENSOR HARDWARE

I²C Sensors

Direct or muxed access
Sensor-specific registers

I²C

Optional I²C Muxes

Channel selection
Shared bus expansion
I²C: SDA / SCL

02 ARC CONTROLLER • Seeed XIAO ESP32-S3

ESP-IDF / FreeRTOS firmware

I2CBus<N>

Device registry / handles
Mux selection
Register reads and writes

I²C access

Sensor<N>

Task per sensor
Setup / read / loop callbacks
Cached readings and offsets

RS485Comm

Sensor ID registry
Receive state machine
Packet validation / response
Sensor readings / offsets

guards

acquires

implements

uses

LockGuard + Bus Mutex

Protect mux selection
and sensor transfer

RS485InterfaceSensor

getID() / getReadings()
setOffsets()

03 ROBOT INTERFACE

RS-485 Transceiver

UART1: TX / RX / RTS
Physical interface

RS-485
half-duplex

V5 Brain

Requests / configuration
Receives sensor data

Connection key Solid line: communication path • Dashed arrow: software dependency • Hollow triangle: interface implementation

CDR design view • Based on ARC source • RS485 communication tested on hardware across multiple packet formats; end-to-end sensor integration remains in progress

https://github.com/Benjamin-Elmore/ARC